

There are three more internal components that need to be understood—Docker images, Docker registries and Docker containers. Docker images are read-only templates; these images contain the operating system with the required libraries/tools and the applications. Docker registries hold these images; the registries are like public/private repositories. The public Docker registry is called Docker Hub and has a huge collection of images with various applications embedded. A Docker container holds everything that is needed for an application to execute; containers are created from a Docker image. Similar to virtual machines, containers can be started, stopped, moved and deleted. Each container provides an isolated and secure environment for applications to run. When Docker runs a container from an image, it adds a read-write layer to the image by using the Union file system.

The Docker registries contain many base images either created by Docker Inc. or the community. These base images are used to build custom images with applications injected. Docker images can be built using a recipe file called Dockerfile.

Now, let's get our hands dirty with Docker. The first step is to install the Docker engine on the host OS or in a VM. Docker is easy to install and it runs on a variety of Linux flavours. It can be installed from the precompiled packages available with the respective Linux distributions or can be compiled from the Docker source available at GitHub. For this article, CentOS 7 x86_64 bit has been installed on a VM with a 4 vCPU (virtual CPU), 16GB memory, a 1GB network port and 100GB of disk space.

Given below are the commands to install Docker on a freshly installed CentOS 7 x86_64 bit version. Please note that the EPEL (Extra Packages for Enterprise Linux) repository needs to be configured for installing Docker packages on CentOS 6.5. However, for CentOS 7, the Docker packages are already available in the default repo. For this article, we disabled the SELinux and FirewallD services before starting with Docker deployment. As we are trying to pull the Docker-relevant packages from the repo, it is imperative to provide proxy details for the machines behind the proxy.

Installing Docker related files through Yum

Type the following commands to install Docker related files through Yum:

```
[root@teja-dev-vm01 ~]# more /etc/redhat-release
CentOS Linux release 7.0.1406 (Core)
[root@teja-dev-vm01 ~]# uname -a
Linux teja-dev-vm01.heLab.bdc 3.10.0-123.el7.x86_64 #1 SMP
Mon Jun 30 12:09:22 UTC 2014 x86_64 x86_64 x86_64 GNU/Linux
[root@teja-dev-vm01 ~]#
[root@teja-dev-vm01 ~]# echo ${http_proxy,ftp_proxy,https_
proxy}
```

```
http://10.10.17.17:3128
[root@teja-dev-vm01 ~]# yum install docker -y
[root@teja-dev-vm01 ~]#
```

After successfully installing Docker, also install *device-mapper-event-libs*. Start the Docker service with the *systemctl* command and also enable the Docker service. Given below are two commands to start the Docker service and enable it:

```
[root@teja-dev-vm01 ~]# systemctl start docker.service
[root@teja-dev-vm01 ~]#
[root@teja-dev-vm01 ~]# systemctl enable docker.service
```

Verify the Docker version installed with the command *docker -v*. The version that got installed was v1.5. With this, Docker installation is complete. The next step is to create a user account on Docker Hub and use those credentials to pull the latest CentOS images.

Browse hub.docker.com and complete the registration process. Once the registration is done, use the command *docker login* to log in to the hub. Given below are the commands to log into Docker Hub for downloading the images. If systems are behind the proxy, then ensure that *HTTP_PROXY* and *HTTPS_PROXY* are defined in the configuration file */etc/sysconfig/docker*.

Log into Docker hub or Docker registry

Here are the commands for log into docker hub or docker registry :

```
[root@teja-dev-vm01 ~]# docker login
Username: my_docker_hub_login
Password: *****
Email : my@email.com
Login succeeded.
[root@teja-dev-vm01 ~]#
```

Once the login to the registry is successful, docker images can be searched using the command *docker search*, images can be pulled from registry, containers can be created from the images, updated images can be pushed to registry, custom images can be built and so many other things can be done. In the commands given below, CentOS images are pulled from repository and then the container is started.

As a next step, let's start the applications inside the container with the fresh CentOS pulled from Docker Hub. The command *Docker run* helps to launch the container with the application. In the following example, the container was launched in interactive mode by using an option *-i*. The option *-t* assigns a pseudo-tty or terminal inside the new container. Along with the above options, specify that */bin/bash* needs to run in the container.

```
[root@teja-dev-vm01 ~]# docker pull centos
Pulling repository centos
```